

Timo Menzel

Graphics Engineer | 3D Reconstruction & Geometry Processing

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[LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

SUMMARY

Researcher and Engineer specializing in 3D reconstruction and geometry processing. Focused on implementing high-performance C++ pipelines that translate 3D computer vision methods into scalable, production-ready systems.

EDUCATION

TU Dortmund University

PhD in Computer Science (Dr. rer. nat.)

- Thesis submitted; currently awaiting defense (Expected Aug. 2026).

Dortmund, Germany

May 2021 – Present

Bielefeld University

M.Sc. in Intelligent Systems

Bielefeld, Germany

Oct. 2018 – Oct. 2020

Bielefeld University

B.Sc. in Informatics for the Natural Sciences

Bielefeld, Germany

Oct. 2015 – Oct. 2018

EXPERIENCE

Research Assistant

Graphics & Geometry Group, Bielefeld University → TU Dortmund University

May 2020 – Apr. 2026

Dortmund, Germany

- Started research position at Bielefeld University; relocated with the group to TU Dortmund in Oct. 2020.
- Bridged the gap between high-level research and production by creating consumer-accessible software for 3D reconstruction.
- Interdisciplinary collaboration to evaluate method performance and **user experience (UX)**.
- Published research findings in high-impact, **peer-reviewed journals** and international conferences.

SELECTED PROJECTS

Avatars for the Masses | C++, Python, Swift, LiDAR, OpenCV, OpenGL, MediaPipe, Eigen

2023 – 2025

- System Architecture:** Developed a scalable server-pipeline for 3D reconstruction, integrating segmentation, photogrammetry, and template-fitting for VR-ready avatars.
- Mobile Development:** Engineered an iOS app for guided RGB and LiDAR data capture.
- Error Compensation:** Extended the pipeline with algorithms to mitigate motion-induced artifacts, improving reconstruction fidelity in uncontrolled environments.
- Visualization:** Integrated reconstruction results into a Unity-based application for real-time rendering.
- Publications:** Main paper in *Frontiers in VR (2025)*; Contribution on error compensation in *ACM VRST 2025*.
- Project Link:** avatars.cs.tu-dortmund.de

Automated Blendshape Personalization | C++, Swift, ARKit, Eigen

2021 – 2022

- Facial Animation:** Developed an automated pipeline for personalizing facial blendshapes from smartphone data to achieve faithful facial expressions.
- Integration:** Utilized ARKit for data acquisition and C++ for backend geometry processing.
- Publication:** Appeared in *ACM VRST 2022*.

Deformation Transfer Demo | C++, OpenGL/Viewer, Eigen

2026

- Mesh Deformation:** Implemented a C++ demo to transfer recorded facial expressions between source and target meshes using deformation transfer.
- Visualization:** Developed an interactive viewer for side-by-side mesh comparison.
- Open Source:** [GitHub Repository](#) | [Interactive Demo](#)

SKILLS

Programming: C++, Swift, Python, C# (Unity), Bash

Computer Vision & 3D: OpenCV, ICP, Photogrammetry, Non-rigid Reg., Halfedge Meshes, ARKit, MediaPipe

Graphics & Math: Linear Algebra, Numerical Optimization, Camera Geometry, OpenGL, Eigen/MKL, PMP-Library

Engineering: Object-Oriented Programming (OOP), Git, CI/CD, Docker, Linux, CMake, GoogleTest

Languages: German (Native), English (Fluent), French (Basic)

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PUBLICATIONS

* Joint first authorship / Equal contribution

- [1] Gambietz, M., Azam, P. Q., Amon, P., Wechsler, I., Hille, E. M., **Menzel, T.**, Ott, T., Botsch, M., Braun, M., Miehl, J., McMahon, K. L., Koelewijn, A. D. “From Body Hulls to Musculoskeletal Models: Personalized Inertial Parameter Estimation”. In: *PLOS ONE* 21.6 (2026), pp. 1–18. DOI: [10.1371/journal.pone.0349886](https://doi.org/10.1371/journal.pone.0349886).
- [2] Runte, F.*, **Menzel, T.***, Schwanecke, U., Botsch, M. “Compensating Motion-Induced Errors in Smartphone-Based VR Avatar Reconstruction”. In: *Proceedings of the 31st ACM Symposium on Virtual Reality Software and Technology*. 2025. DOI: [10.1145/3756884.3765995](https://doi.org/10.1145/3756884.3765995).
- [3] **Menzel, T.***, Wolf, E.*, Wenninger, S., Spinczyk, N., Holderrieth, L., Wienrich, C., Schwanecke, U., Latoschik, M. E., Botsch, M. “Avatars for the Masses: Smartphone-Based Reconstruction of Humans for Virtual Reality”. In: *Frontiers in Virtual Reality* 6 (2025), p. 1583474. DOI: [10.3389/frvir.2025.1583474](https://doi.org/10.3389/frvir.2025.1583474).
- [4] Kullmann, P., Schell, T., **Menzel, T.**, Botsch, M., Latoschik, M. E. “Coverage of Facial Expressions and Its Effects on Avatar Embodiment, Self-Identification, and Uncanniness”. In: *IEEE Transactions on Visualization and Computer Graphics* 31.5 (2025), pp. 3613–3622. DOI: [10.1109/TVCG.2025.3549887](https://doi.org/10.1109/TVCG.2025.3549887).
- [5] Kullmann, P., **Menzel, T.**, Botsch, M., Latoschik, M. E. “An Evaluation of Other-Avatar Facial Animation Methods for Social VR”. In: *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 2023. DOI: [10.1145/3544549.3585617](https://doi.org/10.1145/3544549.3585617).
- [6] **Menzel, T.**, Botsch, M., Latoschik, M. E. “Automated Blendshape Personalization for Faithful Face Animations Using Commodity Smartphones”. In: *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology*. 2022. DOI: [10.1145/3562939.3565622](https://doi.org/10.1145/3562939.3565622).